

messages if it is not connected, as SignalR keeps messages in memory only for 30 seconds. This can be taken care of by resetting the default value of *Disconnect Timeout* in *Application_Start* in your *Global.asax* file.

Other real-time connectivity platforms

PubNub: This is a cloud hosted publish/subscribe messaging service based on the HTTP protocol, implemented in C code. PubNub lets you send or receive millions of messages in milliseconds.

Pusher: This is a simple hosted API for quickly, easily and securely adding real-time bi-directional functionality via Web sockets to Web and mobile apps, or any other Internet connected device.

IGATE's SignalR proof of concept (POC) for SmartMirror

IGATE has developed the SignalR based SmartMirror mobile application that works in conjunction with the SmartMirror client application. The SmartMirror desktop application is a virtual trial room, where users can try clothes without actually wearing them using Kinect technology. Kinect lets the user communicate with the help of gestures. A new user sometimes finds it tedious to try out clothes using gestures, and is more tuned to using the touch interface in a mobile app. So the SmartMirror mobile Web application allows users to control the SmartMirror desktop application using their mobile as the control device, instead of making hand gestures. SignalR has been used as the connectivity platform that integrates the mobile Web application with the SmartMirror client application in real-time.

In the SignalR solution we have a hub/server that is a self-hosted window service. Both the SmartMirror mobile application and the SmartMirror desktop application act as clients to the hub. Now, when the SmartMirror mobile application sends any message to the hub, the latter broadcasts this message to all the clients connected to the hub. Thus, the SmartMirror desktop application can easily be monitored with the help of a mobile application.

1. JavaScript on the client side establishes the connection with the hub. The line of code is:

```
$connection.hub.url = http://myhub/signalr
```

...where *myhub* is the name of the server hosting the hub program.

2. Clients invoke a function on the server when they want to send messages or requests to the hub. If the called function does not match with the function present in the hub, then no response comes from the hub.
3. The hub on return invokes a message on the client as a response to the client's request, and then sends the message to all connected clients without the clients polling for data. It automatically decides the transport

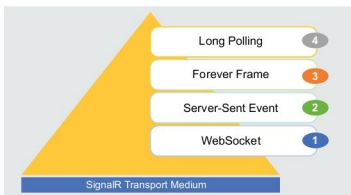


Figure 4: Transport mechanism

medium to opt for, depending upon browser configuration.

IGATE's SignalR POC for digital signage

IGATE has developed a mobile application for controlling digital signage applications. This uses the mvc4 mobile template that has been designed to work with digital signage so one does not have to use gestures; instead, it can be operated via the mobile application. In this POC, SignalR has been used as the connectivity platform for interaction between the mobile application and the desktop application. This solution has a hub/server that is a self-hosted window service. Both the mobile application and the digital signage desktop application act as clients to the hub. Now, when the mobile application sends any message to the hub, the hub broadcasts that message to all clients that are connected to the hub. Thus digital signage can easily be monitored with the help of the mobile application.

How SignalR can be useful in other areas

SignalR provides insight to all real-time Web applications, making them more flexible for users.

- It can be used in communication with touchscreen kiosks.
- It can be used in real-time monitoring applications.
- It can also be used in online gaming applications.
- It can be used in collaborative tools.
- It can be used in job progress updates. 

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